



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (COMPUTER SCIENCE & ENGINEERING / INFORMATION TECHNOLOGY) EXAMINATIONS - JAN/FEB 2023

SUBJECT : MAT 2155 - ENGG. MATHS III
(MAKEUP)

ENGINEERING MATHEMATICS III [MAT 2155]

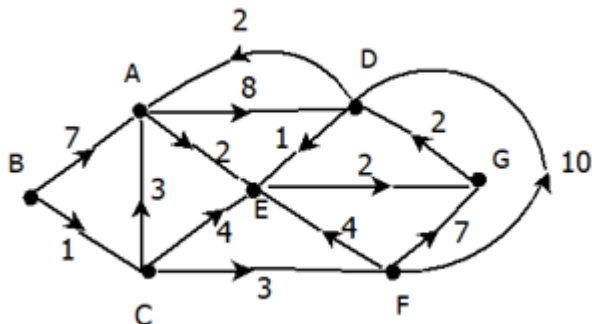
Marks: 50

Duration: 180 mins.

A

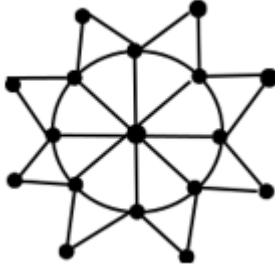
Answer all the questions.

- 1A) Find the 27th and 65th permutation of 5 marks 1,2,3,4,5 in both Lexicographical ordering and Fike's ordering. (4)
- 1B) If 5 men A, B, C, D, E intend to speak at a meeting, in how many orders can they do so without B speaking before A? How many orders are there in which A speaks immediately before B? (3)
- 1C) Prove that in a distributive lattice, if $b \wedge \bar{c} = 0$ then $b \leq c$. (3)
- 2A) i) Let x and y are elements of a group $(G, *)$ and there exist $a \in G$ such that $a * x = a * y$, then prove that $x = y$. (4)
- ii) Prove that in a group $(G, *)$, $(x * y)^{-1} = y^{-1} * x^{-1} \forall x, y \in G$.
- 2B) Prove that Every tree has a centre consisting of either one vertex or two adjacent vertices. (3)
- 2C) Let $(A, \leq)$ be a distributive lattice. Show that $a \wedge x = a \wedge y$ and $a \vee x = a \vee y$ for some a , then $x = y$. (3)
- 3A) Using Dijkstra's algorithm, find the shortest path from B to all other vertices for the network given below. (4)



- 3B) How many ways are there to collect 30\$ from 4 children and 6 adults, if each person gives at least 1\$, but each child can give at most 5\$ and each adult at most 7\$. (3)
- 3C) Define Regular graph, Hamiltonian graph and Eulerian graph. Is the following graph Hamiltonian or Eulerian? Justify your (3)

answer.



- 4A) Using Ferrers diagrams, show that the number of partitions of n into at most k parts is equal to the number of partitions of n with no part larger than k . Hence derive the generating function for the number of partitions of n into at most k parts. (4)
- 4B) Let $E(x_1, x_2, x_3) = \overline{(x_1 \vee x_2) \vee (\bar{x}_1 \wedge x_3)}$ be a Boolean expression over the two-valued Boolean algebra $\{0, 1\}$. Write $E(x_1, x_2, x_3)$ in both CNF and DNF. (3)
- 4C) For any a, b, c, d in a lattice $(A, \leq)$ if $a \leq b$ and $c \leq d$, then show that $a \vee c \leq b \vee d$ and $a \wedge c \leq b \wedge d$. (3)
- 5A) Show that $Z_5 = \{0, 1, 2, 3, 4\}$ forms a cyclic group under the operation of addition modulo 5. List all the elements of Z_5 that generate it. (4)
- 5B) Show that $((p \vee q) \wedge \neg[(\neg p) \wedge (\neg q \vee \neg r)]) \vee (\neg p \wedge \neg q) \vee (\neg p \wedge \neg r)$ is a tautology without constructing truth table. (3)
- 5C) Show that from: (a) $(\forall x)[F(x) \wedge S(x)] \rightarrow \forall y[M(y) \rightarrow W(y)]$
(b) $(\exists y)[M(y) \wedge \neg W(y)]$ the conclusion $(\forall x)[F(x) \rightarrow \neg S(x)]$ follows. (3)

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